

## Blockchain Basics

### Learning Objectives

- Explain the fundamental principles of blockchain technology
- Describe how cryptographic hashing ensures data integrity and security
- Evaluate blockchain applications beyond cryptocurrency

### Engage (5-7 minutes)

Present a puzzle: students form a chain where each person writes a secret number on a piece of paper, folds it, and passes it to the next. Each person adds their own number. To verify the chain hasn't been tampered with, everyone must share their numbers—revealing any tampering. This physical analogy illustrates blockchain's core principle: distributed verification makes tampering evident.

### Explore (10-12 minutes)

Students simulate a simple blockchain manually: each student has a “block” containing a transaction (e.g., “Alice pays Bob 5 coins”), a timestamp, and the hash of the previous block. They compute simple hashes (e.g., sum of ASCII values) and chain blocks together. Then one student tries to alter a past transaction—and the class discovers the tampering because subsequent hashes no longer match.

### Explain (10-12 minutes)

Blockchain is a distributed ledger technology that records transactions across a network of computers. Each block contains: transaction data, a timestamp, and the cryptographic hash of the previous block—creating an immutable chain. Changing any block invalidates all subsequent hashes, making tampering immediately detectable. Key properties: decentralization (no single point of control or failure), transparency (all participants can verify transactions), immutability (once recorded, data cannot be altered), and security (cryptographic hashing and consensus mechanisms). Originally designed for Bitcoin, blockchain now serves many purposes.

### Elaborate (8-10 minutes)

Examine consensus mechanisms: Proof of Work (energy-intensive but secure, used by Bitcoin), Proof of Stake (energy-efficient, used by Ethereum 2.0), and Practical Byzantine Fault Tolerance (fast, used in private blockchains). Analyze blockchain applications: supply chain tracking (Walmart tracing food origin), healthcare records (secure patient data sharing), voting systems (tamper-proof elections), and smart contracts (self-executing agreements). Discuss limitations: scalability issues, energy consumption, and the “blockchain is not always the answer” principle.

### **Evaluate (5-8 minutes)**

Students evaluate whether blockchain is appropriate for three scenarios: a school certificate verification system, a social media platform, and a supply chain tracker. For each, they analyze: need for decentralization, data immutability requirement, transaction speed needs, and cost-benefit. Written justification for each recommendation.